

A Survey on Redactable Blockchain: Challenges and Opportunities

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Abstract—While immutability is an important property of blockchain and can be utilized to build a credible data publishing platform, the same property can also be abused to disseminate illicit content and may violate requirements under the General Data Protection Regulation (GDPR) as well as other privacy regulations. Hence, there has been increased focus on designing redactable blockchain-based solutions, which is the focus of this survey article. Specifically, we survey the extant literature on redactable blockchain, and discuss the existing limitations and challenges prior to outlining a number of future research opportunities which hopefully will benefit the broader research and practitioner communities.

Index Terms—Chameleon hash function, GDPR, immutability, redactable blockchain.

I. INTRODUCTION

WHILE blockchain is commonly associated with cryptocurrencies, especially Bitcoins, its usage and diversity have increased significantly in recent years (e.g., co-existence

of multiple blockchains involving private blockchain and consortium blockchain with access mechanisms). For example, Alibaba introduced Antchain in 2016 to address trust-related challenges in more than 50 scenarios by linking various industrial networks [1], and Citigroup launched the blockchain-based open financing platform for digitizing and sharing related documents in real-time [2].

Decentralization and immutability are two key properties that enable blockchain to ensure trust in an environment involving multiple stakeholders, say governments, banks and other financial institutions [3], [4], [5] (generally denoted as trusted third parties in the traditional centralized transaction system, where transactions are uniformly verified and confirmed by these trusted third parties). The blockchain system eliminates the requirement for trust between users in the decentralized and untrusted environment (e.g., zero-trust setting), since all nodes in the blockchain use multi-party consensus to record and confirm transactions. Immutability also ensures the integrity and traceability of data on the chain; thus, making the audit process more efficient and transparent since an application's complete history and data trail are recorded and can be made readily available.

There are, however, ways the immutability property can create issues. For example, there is no auditing mechanism for the mining process since miners can only check the format of transactions. Hence, illicit content (e.g., child exploitation / abuse materials - also referred to as child pornography) may be stored on and shared via blockchain-based systems [6], [7]. This can create implications on other users on the blockchain, as data from other innocent users (not involved in the illicit activities) may be collected and forensically investigated during a criminal investigation. Clearly, this can create compliance challenges particularly with exacting requirements mandated by the General Data Protection Regulation (GDPR) [8] and other jurisdictional-specific privacy regulations (e.g., the Personal Information Protection Law (PIPL) [9] of the People's Republic of China, introduced in 2021). In addition, GDPR stipulates that users have the "right to be forgotten" (RtbF), where data subjects shall have the right to obtain from the controller the erasure of personal data concerning him or her without undue delay. There are also similar requirements in the PIPL (and potentially other jurisdictional-specific privacy regulations).

From an implementation perspective, the immutability property implies that incorrect data and code recorded on the blockchain cannot be modified. Any incorrect or improper

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TABLE I
A COMPARISON OF CONTRIBUTION BETWEEN RELEVANT SURVEYS
AND OUR SURVEY

	Contribution
[15]	Discussion of blockchain immutability and the RtbF controversy, and various methods of circumventing and eliminating immutability.
[16]	Survey on classic redactable methods.
[17]	A systematic literature review on GDPR compliant blockchains and research domains where GDPR compliant blockchains have been discussed.
Our work	A comprehensive survey of redactable blockchains, including an overview of each approach, as well as a description of their strengths, shortcomings and subsequent improvements, discussions on challenges and potential research directions.

data is not necessarily human error, since such issues can also be caused by the software that interfaces with the underlying database [10]. For example in the context of smart contract, smart contracts can never be modified or updated once they are deployed on the blockchain [11]. Therefore, if there are vulnerabilities in the contract code, these vulnerabilities can be exploited to facilitate attacks such as the DAO [12] (recursive call problem in contract led to 3.6 million ether being stolen) and king of ether throne [13] (the attack caused the contract to fail to execute properly). In addition, the permanent storage of every transaction on the blockchain has cost and performance implications. For example, as of October 2021, the size of the Bitcoin ledger has reached 360 GB [14]. Hence, there has been increased attention from the blockchain community to design redactable blockchain-based solutions.

As shown in Table. I, some review articles [15], [16] emphasize an overview of the types of redacting operations, others [17] focus on GDPR provisions related to blockchain in their classification. The above existing surveys mainly focus on only a representative number of methods. In comparison to them, this article gives a comprehensive survey of challenges and opportunities in redactable blockchain. We focus more on the cryptographic techniques used in the redacting methods in our classification. Then we present each work and its subsequent improvements, so as to help readers better understand the strengths, weaknesses and research lineage of each redacting method.

We introduce several cryptographic techniques that have been used in redactable blockchains (see Section II), as well as studying and categorizing existing methodologies into cryptography-based redaction (see Section III) and non-cryptography-based redaction approaches (see Section IV). For each type of method, we explain its main ideas and characteristics. We then present a comparative summary of existing approaches, before discussing existing limitations and challenges (see Section V). Building on this analysis, we identify a number of potential research directions for future work on

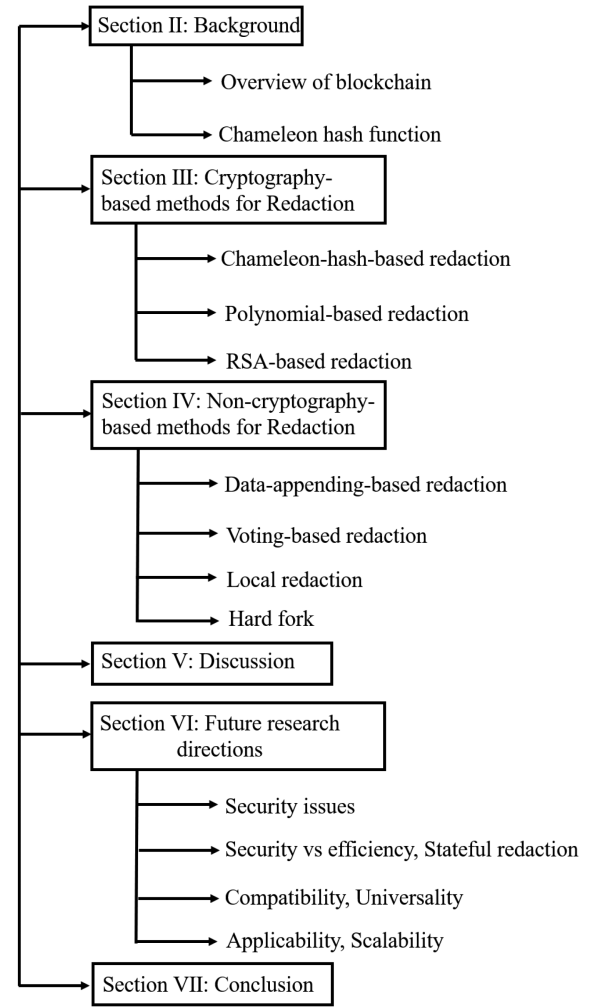


Fig. 1. Organization structure of this paper.

redactable blockchains (see Section VI). Fig. 1 illustrates the organization of this survey.

In the next section, we will briefly introduce blockchain and some cryptographic techniques for completeness.

II. BACKGROUND

A. Overview of Blockchain

Blockchain is essentially a decentralized database, where all nodes jointly maintain the data stored on the blockchain (also referred to as the chain). The blockchain has the following building blocks: blockchain structure (used to store data), distributed consensus algorithm (used to synchronize the data of each node), and cryptographic algorithms (used to ensure security of the account and transaction).

As shown in Fig. 2, the blockchain consists of a set of blocks connected in chronological order and each block mainly consists of the following contents:

- Transaction: The state changes in the blockchain are all accomplished by transactions, where each transaction uniquely corresponds to a transaction hash. In Bitcoin, transactions are processed by scripts [18], and Table II

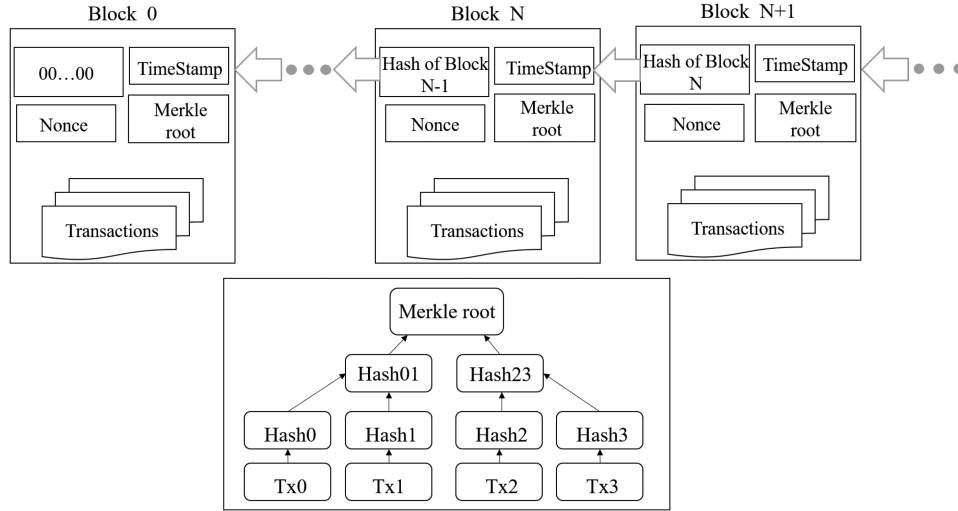


Fig. 2. The basic structure of blockchain.

TABLE II
SCRIPTS IN BITCOIN

	Locking Script	Unlocking Script
P2PK	<Public Key>OP_CHECKSIG	<Signature>
P2PKH	OP_DUP OP_HASH160 <Public Key Hash> OP_EQUALVERIFY OP_CHECKSIG	<Signature><Public Key>
OP_RETURN	–	OP_RETURN <Data>
P2SH	OP_HASH160 <RedeemScriptHash>OP_EQUAL	<Data><RedeemScript>

shows the structure of the scripts in bitcoin. When a bitcoin transaction is validated, the unlocking script in each input is executed along with its corresponding locking script to determine if the transaction meets the payment conditions.

- **Merkle root [19]:** The merkle root value is calculated through all transaction hashes.
- **Previous hash:** This field records the hash value of the previous block. It is designed to achieve the link between blocks.
- **Nonce:** The nonce is computed by miners and used to implement the mining mechanism.

The block hash is calculated by the merkle root value, the previous hash and the nonce. In this article, we use the concept proposed by Garay et al. [20] to describe the blockchain. The $block_i$ is defined as a tuple $B_i := \langle s_i, x_i, ctr_i \rangle$, where s_i is the previous hash value, x_i is the merkle root value and ctr_i is the nonce. N denotes the current block height, C denotes the current blockchain, $B_{N+1} := \langle s_{N+1}, x_{N+1}, ctr_{N+1} \rangle$ denotes the next block, where $s_{N+1} = H(ctr_N, G(s_N, x_N))$. H and G are collision-resistant functions with the output of fixed-size. They are called outer hash function and inner hash function, respectively. The blockchain can be extended to a longer chain $C' = C || B_{N+1}$ by attaching a new block. For a blockchain C of length n and any $0 < k \leq n$, $C^{[k]}$ denotes the state of the blockchain with the k rightmost blocks are removed, $C^{[k]}$ denotes the state of the blockchain with the k leftmost blocks are removed.

In the blockchain, transactions are recorded by all nodes collectively. The consensus algorithm is responsible for maintaining the consistency of all nodes. It avoids the possibility of being spoofed by a single node. The mainstream consensus mechanisms include the following:

- **Proof of work (PoW) [21]:** PoW is the consensus algorithm adopted in the Bitcoin. Its core idea is that all nodes in the network use the specific information to calculate mathematical puzzles to compete for the accounting rights of the ledger. The average block generation time of Bitcoin is ten minutes, so the difficulty of the mathematical puzzle needs to be adjusted according to the computing power of the entire network. The PoW algorithm has the following two problems. One is that as the number of miner increases, the resources required to obtain Bitcoin increase significantly. The other is that the mining pools with large amounts of computing power can dominate the accounting rights.
- **Proof of stake (PoS) [22]:** In PoS, currency age competition replaces the computing power competition. The currency age is calculated by multiplying the number of tokens owned by the node and the time of token ownership. The rights and interests you own are determined by the currency age. Therefore, PoS solves the problems in the PoW well.
- **Practical byzantine fault tolerance (PBFT) [23]:** Different from the above two algorithms, PBFT is a cooperative

consensus algorithm. There is a primary node and some backup nodes in this algorithm. The primary node sorts the client requests and sends them to the backup nodes which then process the requests. If more than one-third of the nodes approve, the request is approved.

The cryptographic algorithms are used to ensure the security of the account and transaction. The hash function and digital signature are used in the blockchain currently. The hash function has the characteristics of collision resistance and uniformity. The blockchain usually save the transaction information's hash values. The use of the hash function improves the operational efficiency and scalability of the blockchain. Moreover, it eliminates the need to encapsulate all underlying data in the block header. The digital signature is used to verify the messages' origin and integrity. In Bitcoin, the public key of the transaction recipient is used to generate the locking-script. Only the private key owner can create the signature corresponding to the unlocking-script. It ensures the account's security.

Based on the blockchain's architecture described above, we describe the characteristics of the blockchain [24] as follows.

- **Anonymity:** In the blockchain, each node is identified by the public key. The user's identity in the real world is hidden and only the user's pseudonym is available. Therefore, it is impossible to obtain the user's real information in the network.
- **Immutability:** The hash pointer guarantees the immutability of the blockchain. If a block is modified, the hash verification will fail. Furthermore, each submitted transaction needs to be verified and validated by all nodes before it is recorded in the blockchain.
- **Decentralisation:** The blockchain is built on the P2P network. The transactions of digital assets can be done in a secure and decentralized fashion. It makes the system free of governing authority.
- **Transparency:** The data in the blockchain ledger is publicly visible. It can be accessed and verified by any node in the network.

B. Chameleon Hash Function

The chameleon hash (CH) function was first proposed by Krawczyk and Rabin [25]. It is defined as a collection of algorithms in the following.

Definition 1 (Chameleon hash function). A chameleon hash function is a tuple of algorithms $A = (\text{KeyGen}, \text{Hash}, \text{HVer}, \text{HCol})$

- $(pk, tk) \leftarrow \text{KeyGen}(1^k)$: The key generation algorithm takes the security parameter k as input and returns the public key pk and the trapdoor key tk .
- $h \leftarrow \text{Hash}(pk, m, r)$: The hash function takes the public key pk , the message m and the random number r as input and outputs the hash value h .
- $\{0, 1\} \leftarrow \text{HVer}(pk, m, (h, r))$: The hash verification function takes the public key pk , the message m , the random number r and the hash value h as input. If h is

KeyGen: (p, q, g, y, x)

q : a large enough prime factor

p : a large prime and $p = kq + 1$

g : an element of order q in $\mathbb{Z}_p^*$

y : the public key and $y = g^x \bmod p$

x : the private key and $x \in \mathbb{Z}_q^*$

Hash: $(y, m; r, h)$

r : a random value and $r \in \mathbb{Z}_q^*$

h : the hash value and $h = g^m y^r \bmod p$

HCol: $(x, m, r, m'; r')$

r' : the random number corresponding to the new message m'

$$g^m y^r = g^{m'} y^{r'} \bmod p$$

$$m + xr = m' + xr' \bmod q$$

$$r' = \frac{m - m'}{x} + r$$

Fig. 3. Chameleon hash based on discrete log assumption.

the hash value corresponding to the message m , output 1, otherwise output 0.

- $r' \leftarrow \text{HCol}(tk, (h, m, r), m')$: The collision finding algorithm takes the trapdoor key tk , the original message m , the original random number r , the hash value h and the new message m' as input, and outputs the random number r' corresponding to the new message m' .

The chameleon hash function has the following properties: anyone with the trapdoor key can easily find the collision of the hash. But for those who do not know the trapdoor key, the hash function is collision resistant. The first construction of chameleon hash function is based on the discrete logarithm assumption (DLP). The algorithm process is shown in Fig. 3. However, Ateniese and Medeiros [26] found the key exposure problem in this construction, i.e., if two pairs (m, r) and (m', r') are exposed the trapdoor key x will be recovered from $h = g^m y^r = g^{m'} y^{r'}$ and $x = \frac{m' - m}{r - r'}$.

The key exposure problem makes the chameleon hash function unsuitable in the redactable blockchain. Therefore, Ateniese et al. [27] proposed the notion of enhanced collision resistance: even if polynomial collisions are exposed, it should be difficult to find collisions for the hash function. They also gave a concrete instantiation. It is based on the twin Nyberg-Rueppel signature [28]. Based on this instantiation, they cooperated with Accenture [29] to develop a prototype on the Hyperledger [30] for permissioned environments, in which trusted entities can redact blockchain according to the corresponding rules.

III. CRYPTOGRAPHY-BASED METHODS FOR REDACTION

This section presents a comprehensive overview of cryptography-based redactable blockchain solutions recently proposed. Those mainstream methods can be mainly divided into: chameleon-hash-based redaction, polynomial-based redaction and RSA-based redaction. Table III provides a summary of these methods.

TABLE III
THE SUMMARY OF REDACTABLE METHODS IN SECTION III

Methods	Works	Main Idea
Chameleon-hash-based redaction	[27], [31], [32] [33], [34], [35], [36]	Replacing the inner function with the chameleon hash function.
Polynomial-based redaction	[37]	The block is represented by a polynomial and every transaction represents a coordinate. Users can change transactions while keeping the polynomial unchanged
RSA-based redaction	[38], [39]	The hardness of the RSA problem makes the blockchain immutable, the user who owns the private key can redact the blockchain.

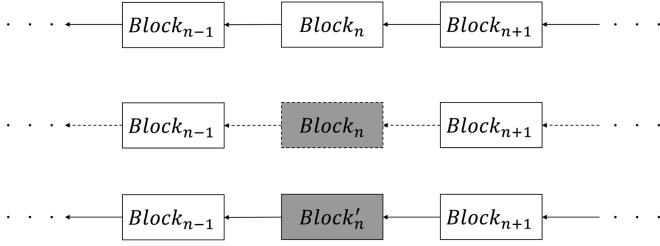


Fig. 4. Redaction operations.

A. Chameleon-Hash-Based Redaction

In the blockchain, the collision resistance property of the hash function makes the blockchain immutable. One of the solutions to redact the blockchain is to adopt the chameleon hash function. This way was first presented by Ateniese et al. [27]. Their main idea is to replace the inner hash G with an enhanced collision-resistance chameleon hash function. When redacting the block, we only need to find the collision of the inner hash without recalculating the proof of work. We present this method below.

As shown in Fig. 4, in the top blockchain, the blocks are linked with the SHA256 algorithm [40] (solid arrow) and they are immutable. In addition, blocks at other layers are linked by using the chameleon hash function. With the knowledge of the trapdoor key (dashed arrow), it is possible to find collisions and redact the blocks efficiently. Therefore, the $Block_n$ can be replaced with $Block'_n$. The blockchain remains immutable (solid arrow) if the trapdoor key is secret. In their scheme, the $block_i$ is defined as a tuple $B_i := \langle s_i, x_i, ctr_i, r_i \rangle$, where r_i is the parameter for the chameleon hash function, and others are defined as in Section II. In the following, we let $h_i = Hash(pk, s_i || x_i, r_i)$. Two chain redacting algorithms are defined to rewrite and delete blocks.

The inputs of the rewriting algorithm are the chain C , the index i of the redacted block, the new value x' of $block_i$ and the chameleon hash trapdoor key tk . Meanwhile, it uses the new content to compute collision for the hash of $block_i$. A new chain C' can be constructed by replacing $block_i$ with $block'_i$. Then, the new chain is broadcast on the network. According to the redacting rule, every user should adopt the new chain, even if they store a longer one. Those specific steps are as follows:

```

Parse the chain  $C$  as  $B_1 B_n$ 
for  $j = 1$  to  $n$  do
  if  $j == i$  do
    Parse  $B_i = \langle s_i, x_i, ctr_i, r_i \rangle$ 
     $r'_i \leftarrow HCol(tk, (h_i, s_i || x_i, r_i), (s_i || x'_i))$ 
     $B'_i := \langle s_i, x'_i, ctr_i, r'_i \rangle$ 
     $C \leftarrow C^{[n-i+1]} || B'_i || C$ 
  end if
end for
return  $C$ 

```

Assuming that B_i will be deleted, the deleting algorithm takes the chain C , the index i and the trapdoor key tk as inputs. What we need to do is replacing the pre-hash value s_{i+1} of B_{i+1} with the B_i 's pre-hash value s_i . Then, we can compute $r'_{i+1} = HCol(tk, (h_{i+1}, s_{i+1} || x_{i+1}, r_{i+1}), (s_i || x_{i+1}))$ and broadcast the chain $C = C^{[n-i+1]} || B'_{i+1} || C$ on the network.

In this scheme, any redacting operations do not change the block hash. Therefore the consistency of the block is guaranteed. The critical problem is how to manage the trapdoor key. Because the security of the scheme relies on the trapdoor, once the trapdoor is exposed, the redacting rights will be abused. According to the type of blockchain, trapdoor management solutions can be divided into three types. In the private blockchain, the trapdoor key could be given to the central authority. In the consortium blockchain and public blockchain, the trapdoor key could be shared among parties of the blockchain, and redacting operation is completed by using multi-party computation (MPC) [41] and key sharing [42].

However, the above scheme has three problems: (1) Because redaction is achieved on the block-level, it is too coarse-grained to manage redaction privileges conveniently. (2) When a block is rewritten or deleted, the old state of the block will not be retained. It will make public accountability and verifiability impossible. (3) There are key management problems in the scheme. Because the MPC protocol is unsuitable for the situation where there are thousands of users in the blockchain.

Derler et al. [31] proposed the policy-based chameleon hash (PCH) to solve the first problem. It combines access control and chameleon hash. Unlike Ateniese et al.'s scheme, the chameleon hash function in their scheme is used for transaction hash, so changing the transaction will not affect the value of the merkle root. Moreover, users are tagged by some attributes and can tag the transaction with an access policy before integrating it

into the blockchain. Only the user who satisfies the access policy can redact the blockchain. This scheme enables fine-grained redaction. When performing redaction, only the corresponding transaction needs to be operated rather than the block. Moreover, it solves the key management problem through the ciphertext-policy attribute-based encryption scheme (CP-ABE).

However, this scheme also lacks the function of accountability and verifiability. Because only the latest state of data is stored on the blockchain. What is more, the policy may suffer from some attacks. On the one hand, the security of this scheme depends on the attribute policy set by the user. If an attacker's attributes can be satisfied, the policy may be useless. On the other hand, there is a central authority in the CP-ABE scheme. The central authority may be mischievous, e.g., authorizing illegal users. To solve the authorization problem, Ma et al. [36] proposed the decentralized policy-based chameleon hash (DPCH). Compared to the PCH, multi-authority attribute-based encryption is used in this scheme and the rewriting privilege is issued by multiple authorities.

To solve the second problem, Huang et al. [32] proposed the redactable consortium blockchain (RCB), which combines the threshold chameleon hash (TCH) and accountable-and-sanitizable chameleon signature (ASCS). In this scheme, all operations are performed by sensors. The user sensors generate transactions by ASCS. The authorized sensors are authorized to maintain the RCB and the trapdoor key tk is secretly shared among them. When receiving a redacting request, they can compute the hash collision for the block with the TCH and update the signature with ASCS.

RCB effectively addresses corruption in the redacting process through multi-party key management. Meanwhile, it achieves the accountability function. When a transaction is in dispute, users can reveal the transaction's origin by providing a collision of the signature. However, RCB suffers from these drawbacks. First, the scheme is inefficient because it requires the cooperation of authorized sensors to calculate the collision. Second, the RCB is designed only for industrial IoT and is not applicable to other environments.

Another way to address the second problem is to punish these malicious behaviors through the central authority (CA) (Xu et al.) [33]. They proposed the k -time modifiable and epoch-based redactable blockchain (KERB). Before redacting, the modifier must make a time-locked deposit (the deposit can be extracted by the CA and modifier's signature before the expiration time) and send a key issuing request to the CA to obtain the redactable privilege. In order to reduce malicious behavior, the rewrite authority defined by CA is limited to k times at most. If the modifier has made more than k -time redactions, the random number in the signature will be reused. In that case, CA can reveal the secret key of that modifier and get the deposit with CA's signature and that of the modifier before the expiration time.

KERB achieves accountability mechanisms through deposit locking and CA authorization. Moreover, k -time redaction effectively solves the permission revocation problem. What is more, the scheme improves the efficiency of redacting. It applies digital signatures to achieve redacting rather than

complex cryptographic primitives. However, KERB needs a trusted CA to ensure system security. If the CA misbehaves, the whole system will be out of control.

For the third problem, there is no suitable solution currently. Most schemes have adopted the MPC protocol, however it is inefficient when the number of parties becomes large enough. Therefore, sharing trapdoor key tk among a few parties might be a good idea. For example, participants are selected based on their computing power or stake. However, it will influence the system's security and the redaction's fairness, because it is based on a trusted majority. Therefore, we should avoid using the MPC protocol in the implementation process.

Some researchers have made improvements in other aspects. Camenisch et al. [35] proposed the chameleon hash with ephemeral trapdoor keys. In their scheme, a collision could be computed if and only if the trapdoor key and ephemeral key were known. Moreover, they defined a formal security model for the chameleon hash, which presented a more robust definition and could be helpful in other scenarios. Wu et al. [34] introduced a single-trapdoor key-exposure free chameleon hash function based on quantum-resistant assumptions, which is the first construction based on the lattice and does not involve complex cryptographic primitives such as encryption.

In the past few years, chameleon-hash-based redactable blockchains have been integrated into various systems to extend their functionality and performance. Zhang et al. [43] proposed industrial data management system. In this system, a redactable blockchain can reduce the size of industrial data and delete erroneous data. It ensures the security of industrial data and the stability of the devices. Xu et al. [44] introduced an identity management and authentication system. Redactable blockchain effectively solves the problems posed by revocation lists. Compared to previous systems, it reduces storage overhead and prevents illegal user logins. Besides, Mishra et al. [45] designed the redactable-blockchain-based secure data aggregation system. Wei et al. [46] introduced the redactable blockchain framework for secure federated learning.

B. Polynomial-Based Redaction

Cheng et al. [37] proposed the polynomial-based redactable blockchain (PRB). In the polynomial-based architecture, the Lagrange interpolation method organizes the data in every block. A polynomial represents the block. The block header and every transaction are defined separately by coordinates. According to the Lagrangian interpolation formula [47], there is a unique polynomial that satisfies all these coordinates. The order of blocks is kept by recording their polynomial function into their following block header.

The block structure is shown in Fig. 5. In the block header, f^{pre} represents the polynomial of the previous block and info field contains a prime p and the redaction difficulty d , where d is a value randomly selected from 0 to $p - 1$. Padding field is set to be *null* at the beginning. The coordinate is calculated by the following steps: $x^{header} = \text{HASHCUT}(H(f^{pre} \parallel \text{info} \parallel \text{padding}))$, where *HASHCUT* is an algorithm that maps string to an integer. y^{header} is selected from 0 to d . For the

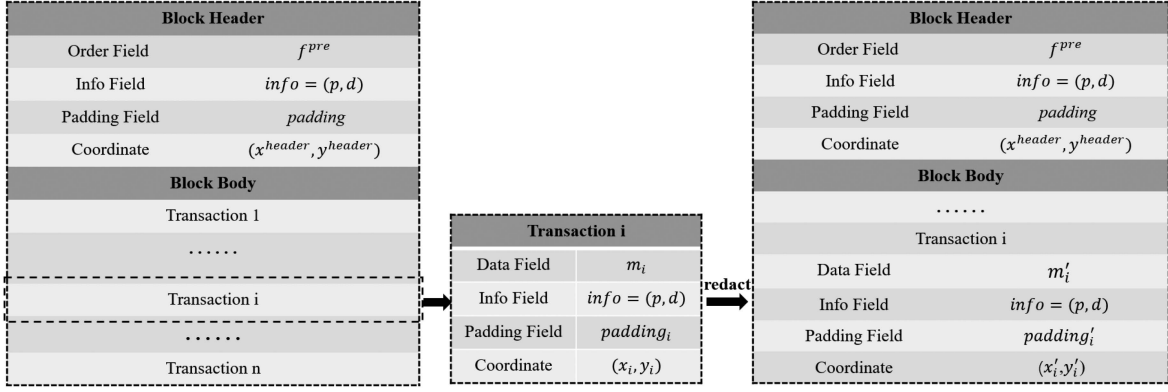


Fig. 5. The structure of polynomial-based blockchain.

Public information:

n : a large integer, which is a product of two large primes
 H : a hash function, e.g. SHA-256

Private information:

p, q : prime factors of n , they are of the form $2r+1$, where r is a prime

Block creation:

x_i : a integer between 1 and $n-1$, and $x_i^2 \neq 1 \pmod n$
 $h_i: h_i = H(p_i, c_i)$
 $d_i: d_i = h_i \bmod n$
 $p_{i+1}: p_{i+1} = x_i^{d_i^2+1} \bmod n$

Block redaction:

$h'_i: h'_i = H(p_i, c'_i)$
 $d'_i: d'_i = h'_i \bmod n$, and $(d_i'^2 + 1)$ should be relatively prime to $\varphi(n)$ by using a padding
 e'_i : the inverse of $(d_i'^2 + 1) \bmod \varphi(n)$
 $x'_i: x'_i = p_{i+1}^{e'_i} \bmod n$

Fig. 6. The implementation of RSA-based blockchain.

transaction, data field contains the transaction message. The information of info field and padding field is the same as in the block header. The coordinate of transaction is calculated by the following steps: $x_i = \text{HASHCUT}(H(m_i \parallel \text{info} \parallel \text{padding}_i))$ and y_i is selected from 0 to d . When performing the redaction operation, assuming that the user redacts the data of transaction_i . First, the user replaces the data field m_i with the new data m'_i and sets the padding field with a random value $\text{padding}'_i$. Then, the new coordinate x'_i is calculated by m'_i and $\text{padding}'_i$, the y'_i is calculated by substituting x'_i into the block's polynomial. If $y'_i \in (0, d)$, let (x'_i, y'_i) be the coordinate of transaction_i . Otherwise, reselect a random value $\text{padding}'_i$ to calculate the coordinate. When deleting and inserting the block, user replaces the order field f^{pre} with the $f^{\text{pre}'}$, and computes the coordinate of block header as the step above.

PRB supports redaction and deletion at low cost, because the whole process does not involve complex operations. Moreover, modification difficulties can be adjusted to suit different security requirements and scenarios. However, there are three main limitations that need to be considered:

1. The historical data will be covered with the most recent one when performing redaction operations. It makes accountability and audit impossible.
2. This scheme is not suitable for the PoW consensus. On the one hand, creating a block does not need too much computing power, and adopting the PoW consensus algorithm will lead to a higher rate of forks. On the other hand, if using the PoW consensus algorithm, the miners will choose the transactions depending on factors such as transaction fees. Some modifications may not be recorded timely.
3. PRB is incompatible with mainstream blockchains, and requires significant changes to the structure of the original blockchain when implemented.

C. RSA-Based Redaction

Grigoriev and Shplrain [38] proposed the RSA-based redactable blockchain. It focuses on the private blockchain. In this construction, the hardness of the RSA problem [48] makes the blockchain immutable.

The block_i is defined as $B_i := \langle p_i, c_i, x_i \rangle$, x_i is the redactable suffix, c_i is the content of block, p_i is the permanent prefix. There are two values $h_i = H(p_i, c_i)$ and $p_{i+1} = F(h_i, x_i)$, where H is a hash function and F is a one-way function. When performing integrity checks, for each block_i it is only necessary to calculate whether $p_i = F(h_{i-1}, x_{i-1})$ holds. Assuming that the user redact the block_i , c_i or p_i is changed, then h_i will also be changed. The main idea is that the central authority keeps the p_{i+1} unchanged by selecting a new suffix x'_i and the private information he owns. The specific construction is shown in Fig. 6. When redacting the block, the redaction steps are as follows:

1. User computes the hash $h'_i = H(p_i, c'_i)$ with the new content c'_i .
2. User converts h'_i to an integer d'_i by using a padding to have $(d_i'^2 + 1)$ relatively prime to $\varphi(n)$.
3. User computes e'_i , which is the inverse of $(d_i'^2 + 1) \pmod{\varphi(n)}$.
4. User computes the new suffix $x'_i = p_{i+1}^{e'_i} \pmod n$. The following equation holds $(x'_i)^{d_i'^2+1} = (p_{i+1}^{e'_i})^{d_i'^2+1} =$

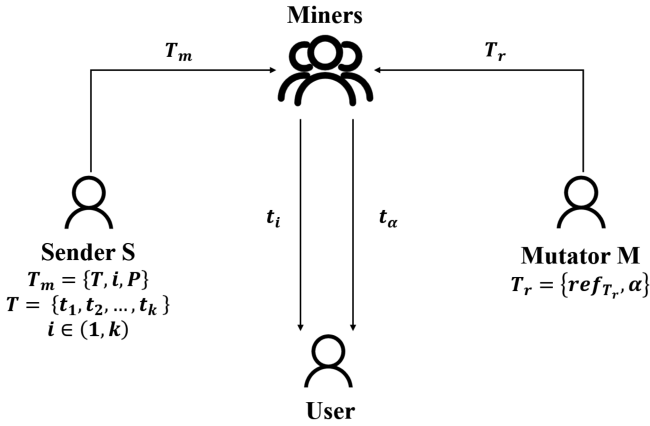


Fig. 7. Redaction process of Puddu et al.'s scheme.

$p_{i+1} \pmod{n}$. Because $e'_i(d_i^{n^2} + 1) = 1 \pmod{\varphi(n)}$ and $(P_{i+1})^{\varphi(n)} = 1 \pmod{n}$.

This scheme provides strong security based on cryptography. On the one hand, we can see from the above construction that if a user who does not know the private information $\varphi(n)$ wants to modify the block, he has to solve the RSA problem. On the other hand, it is corruption-resistant, i.e., replacing the block with something meaningless. The user needs to compute d_i from the random X_i and the equation $P_{i+1} = (X_i)^{d_i^2+1} \pmod{n}$. To do that, the user has to solve the discrete logarithm problem.

However, this construction may suffer from some attacks. Dousti et al. [39] proposed two attacks on this construct. The first is called the moderator circumvention attack. The attacker can generate two blocks, then add one of them to the blockchain, and he can replace it with the other one at any time. It works as follows:

1. Let $h_i = H(p_i, c_i)$ and $h'_i = H(p_i, c'_i)$.
2. Let $d_i = h_i \pmod{n}$ and $d'_i = h'_i \pmod{n}$.
3. Pick x from $\mathbb{Z}_n$, let $x_i = (x)^{d_i^2+1}$ and $x'_i = (x)^{d_i'^2+1}$.
4. $p_i = (x_i)^{d_i^2+1} = (x'_i)^{d_i'^2+1} = p'_i$.

Therefore, replacing blocks does not affect the integrity verification of the blockchain. The second is called the reversion attack. The idea of this attack is as follows, assuming that block B_i is replaced by B'_i , the attacker can revert the version of the block, because all versions can pass verification.

IV. NON-CRYPTOGRAPHY-BASED METHODS FOR REDACTION

This section presents a comprehensive overview of non-cryptography-based redactable blockchain solutions recently proposed. Those mainstream methods can be mainly divided into: data-appending-based redaction, voting-based redaction, local redaction and hard fork.

A. Data-Appending-Based Redaction

Puddu et al. [49] proposed a new protocol $\mu chain$. As shown in Fig. 7, mutable transactions are defined as (T, i, P) in $\mu chain$. T represents a transaction set that contains an empty transaction and other transactions must have the same sender and receiver. i identifies the active transaction in this set. P is

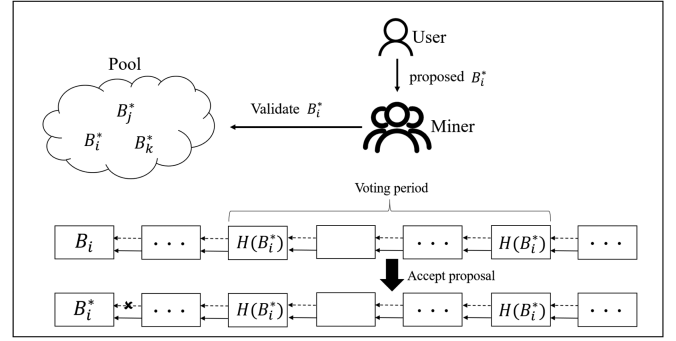


Fig. 8. Redaction process of Deuber's scheme.

the redaction policy, which is specified by the transaction sender and defines all conditions that should be satisfied when redacting the transaction. All transactions in this set will be encrypted and validators will share decryption keys valid for active transactions. When performing redacting operations, the user can send a mutant transaction to change the active transaction in the transaction set. If the transaction set does not contain the desired one, the user can send an extending transaction to extend the set. The empty transaction will be the active transaction in the previous set. Moreover, the user can issue several mutant transactions and extend transactions for the same set. The last one will be specified as the active transaction.

In $\mu chain$, redactions are made by sending transactions so that the consensus mechanism can ensure the security of the blockchain. Moreover, it stores all historical data and can satisfy integrity verification no matter how many times blocks are redacted.

However, there are four problems with this scheme. First, this scheme can not modify the data of the blockchain essentially. It only changes the validators' view of the transaction. Second, The illegal data is still on the blockchain, and users might store the keys of old active transactions locally. Third, a transaction set represents the transaction. It wastes a lot of storage space and affects the scalability of the blockchain. Finally, this scheme is unfeasible because of the high overhead. On the one hand, every transaction is encrypted by different keys. On the other hand, the key management scheme is complicated for those keys.

B. Voting-Based Redaction

This model uses a voting mechanism to realize redaction consensus. There are two different ways to achieve this model. One is to keep two hash links by expanding the block structure. It was first proposed by Deuber et al. [50]. Marsalek and Zefferer [51] introduced another way. The core idea of their scheme is to keep two chains, one holding the original data for validation and the other storing the new data for redacting.

For Deuber et al.'s scheme [50], they add the initial state of the inner hash value y to the block structure. The $block_i$ is defined as $B_i := \langle s_i, x_i, ctr_i, y_i \rangle$, the block hash $h_i = H(ctr_i, G(s_i, x_i), y_i)$. $y_i = G(x_i, s_i)$, which will not be changed after initialization.

The redaction process is shown in Fig. 8. Assuming that a user wants to redact the Block B_i . Firstly, he needs to send a

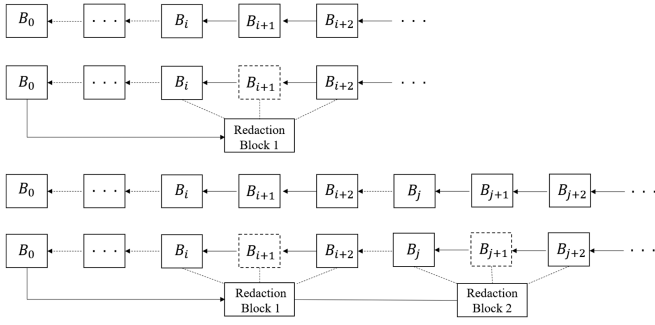


Fig. 9. Redaction process of Marsalek et al.'s scheme.

redacting request in the network. It contains the index i and the candidate block B_i^* . Secondly, if the request is valid, miners will add it to the candidate pool R . Additionally, during the voting period, they can vote for the request by adding the hash of the request in the Coinbase of the block. Finally, after the voting period, everyone can verify whether the request have been approved. If the request gets enough votes, B_i will be replaced by B_i^* . Then, the user broadcasts the new blockchain in the network. There are two hash pointers in the block. The dashed arrow represents the new state $h_i = H(ctr_i, G(s_i, x_i), y_i)$. If x_i or s_i is redacted, B_i 's block hash value h_i and the previous hash value s_{i+1} in B_{i+1} will not be equal. The solid arrow represents the old state $h_i = H(ctr_i, y_i, y_i)$, which is always equal to s_{i+1} and ensures the integrity verification of the blockchain.

This method achieves multi-party redacting through voting. Moreover, it can still ensure accountability even if someone has redacted the blockchain. Because the hash of the initial transaction is stored, the user can provide the removed data to verify this hash. It can prevent data from being redacted maliciously. What is more, the voting protocol is based on the PoW consensus. It is difficult for adversaries to control voting unless they occupy a large percentage of the computing power.

However, this method suffers from two significant drawbacks: 1) It only supports the redaction of data that can never be spent and fails to redact the stateful contents, such as the UTXO set. 2) The scheme is inefficient, because the voting period requires collecting votes in consecutive 1024 blocks. 3) This scheme modifies the structure and protocol of the blockchain, so it is incompatible with mainstream blockchains such as Bitcoin and Ethereum.

For Marsalek and Zefferer's scheme [51], the main contribution is to address the limitation of data redactions. They proposed a new blockchain architecture, including a *Standard Chain* and a *Redaction Chain*. When a user wants to redact a block, the process is similar to that of Deuber et al.'s scheme [50]. It includes the following steps: proposing a request, verifying it, and voting for the proposal by miners. Fig. 9 shows the expansion process of two chains. The redaction will be integrated into the *Redaction Chain* if the request is approved. Otherwise, the *Standard Chain* is extended as usual. The proposal is to remove only the unwanted content from the unspent transactions that might be the input of another transaction. For example, it is unlikely that someone

has the matching private key for P2PK scripts in which the public-key field is used to store data. If they are unspent for a long time, miners can get the money when redacting. This method is also applicable to other Bitcoin scripts. It can reduce the size of the UTXO set and remove unwanted data. However, this scheme is also voted on the blockchain, which is inefficient.

To achieve the instant redaction, Li et al. [52] proposed a new protocol, which is an improvement of Deuber et al.'s scheme [50]. It utilizes a committee to make the redacting decision off the blockchain. The redacting steps are as follows:

1. The user proposes a redaction request and broadcasts the candidate block to the network.
2. Upon receiving the request from the network, every party of the committee verifies the candidate block. If it is valid, they store it in their redacting pool.
3. Permitted parties vote for candidate blocks during the voting period and broadcast their votes with their signatures.
4. The corresponding leader collects and validates the votes on candidate blocks. If enough votes are collected, the leader replaces original blocks with candidate blocks and broadcasts the new chain in the network.

This scheme can achieve instant redaction because it collects votes off the blockchain within a specified time. Compared with the above two schemes [50], [51], voting does not depend on block generation time and the redaction can be as fast as the underlying blockchain. However, this scheme's security relies on committee members' honesty. Therefore, committee members should be randomly selected among all parties according to their stake and computing power.

In addition to the mentioned researches, some other studies have been proposed. Daniel and Tschorsch [53] introduced the notion of permanent and removable blocks. The former contains immutable information. The latter contains the information that can be modified and all redacting operations are performed on the removable blocks. These two blocks are created by selecting different transaction types in their scheme. Xu et al. [54] proposed the redactable proof-of-stack blockchain, which is similar to the above instant redaction scheme [52]. Thyagarajan et al. [55] proposed the Reparo protocol, which can perform repairs as a publicly verifiable layer on the blockchain. In their scheme, all versions of the record are stored in the database, which provides additional functions for the blockchain. Therefore, Reparo can integrate with the current blockchains.

C. Local Redaction

Florian et al. [56] proposed the concept of functionality-preserving local erasure (FPLE), which enables only the local erasure of data. FPLE allows users to mark data for erasure and stores erased data in an erasure database. Meanwhile, FPLE employs the following principles: miners ignore transactions whose input includes the erased data when mining the block. For blockchain growth, FPLE adopts a method similar to the simplified payment verification (SPV) to verify blocks,

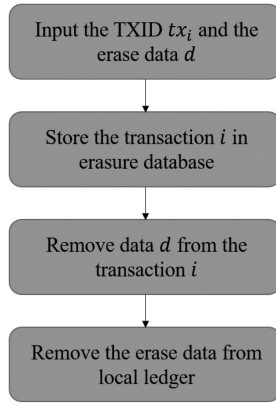


Fig. 10. The erasure process.

i.e., only verify the block headers. Because Bitcoin's incentive mechanism will drive miners to mine effective blocks.

Although the full node stores complete data locally, the erasure request will be discovered only after data analysis and media reporting. The process of erasure is visualized in Fig. 10. First, the user retrieves the related $transaction_i$ through the TXID. Then, the user stores the $transaction_i$ in the erasure database for subsequent operations and removes the erase data d from $transaction_i$ to get $transaction'_i$. Finally, the user replaces $transaction_i$ with the $transaction'_i$ in the local ledger. FPLE records the data being edited through the erasure database, which ensures the consistency in chain growth. Furthermore, this method does not need to modify the structure and protocol of the blockchain, so it is compatible with current blockchains.

However, this scheme has two limitations: (1) Local redaction is personalized, the user can only redact data locally and these data may still be stored in other nodes. (2) FPLE leads to many miners keeping an incomplete ledger. It will endanger the network and limit nodes' functionality.

D. Hard Fork

A hard fork [57] is a radical change to the network's protocol. As shown in Fig. 11, a hard fork will change the blockchain protocol and create two branches. One follows the previous protocol, and another follows the new one. A hard fork can be considered a revocation operation in any blockchain platform. In the DAO attack [12], the purpose of a hard fork is to transfer DAO-related funds to the newly created smart contract and help the token owner get their ether returned.

When a hard fork happens, all miners must agree on the new rules and upgrade the protocol. However, the consensus is not easy to reach in the peer-to-peer network. Moreover, a hard fork cannot be considered a real modification because the previous data is still stored on the old blockchain. What is more, a hard fork requires regenerating the blockchain from a specific block, which consumes a lot of computing power.

In addition to the mentioned methodologies, i.e., data-appending-based redaction, voting-based redaction, local

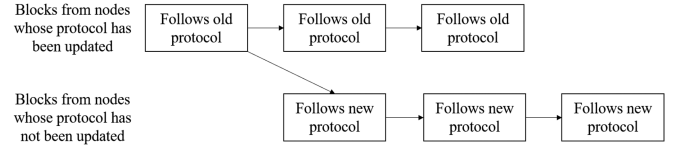


Fig. 11. Hard fork.

redaction and hard fork, there are also some mechanisms that can be used to redact the blockchain. Bruce [58] proposed the Mini-Blockchain scheme, which is a variant of the Bitcoin protocol. In this scheme, the functions of the blockchain are divided into individual mechanisms, so that the user does not need to store the whole blockchain. Matzutt et al. [59] presented a snapshot-based pruning scheme, called CoinPrune. In this scheme, miners announce the recent snapshot of the blockchain. When nodes are synchronized, they only need to obtain the state of the network through snapshots instead of downloading the entire ledger. LEE et al. [60] utilized the sidechains and the truncated hash to achieve redaction. Feng et al. [61] proposed a prunable shared-based protocol which is based on the improved Rollerchain [62] and PBFT algorithm.

V. DISCUSSION

In this section, we summarize and discuss the methodologies described above. We focus our comparison on their impact on blockchain redaction, main disadvantages, and existing implementations.

Table VI summarizes and compares the redacting methodologies in Sections III and IV in terms of some aspects, including their scope of application and main disadvantages. The comparisons show that a few methods support the transaction-level redaction and the redaction of existing data. Most methods are designed for the PoW consensus and provide the function of accountability. Regarding redacting efficiency, most schemes based on chameleon hash are designed with a centralized approach, so they are very efficient when performing redacting. However, due to the long voting period, voting-based schemes are low efficiency. In terms of stateful redaction, almost all schemes do not consider the context of the edited transaction. Therefore, most of them do not support redacting the UTXO set.

Chameleon-hash-based redaction makes it easy to find collisions of hash functions through trapdoor keys. It is the only method that has been implemented in practical projects currently. Table IV compares chameleon-hash-based solutions in Section III-A in terms of their technologies, assumptions, and security model. From the table, we can learn that all of these schemes have strict security proofs and use a variety of cryptographic primitives to ensure the scheme's security. The key management and the construction of the enhanced collision-resistance chameleon hash function play a significant role in guaranteeing the scheme's security. Integrating the chameleon hash function with blockchain is an elegant solution in the private and consortium blockchain where the trapdoor key is shared among a few parties.

TABLE IV
COMPARISON AMONG CURRENT CHAMELEON-HASH-BASED METHODS

Works	Technologies	Assumptions	Security Model	Time
Ateniese et al. [27]	CH	DLP	Generic Group Model	2017
Camenisch et al. [35]	CH	RSA and DLP	Random Oracle Model	2017
Derler et al. [31]	ABE, PCH	–	Strong Indistinguishability	2019
Huang et al. [32]	ASCS, CH	CDH	–	2019
Xu et al. [33]	CH, Signature	–	w-KE Security Model	2021
Wu et al. [34]	CH	ISIS and SIS	Random Oracle Model	2021

*ABE refers to attribute-based encryption.

*CDH refers to Computational Diffie-Hellman assumption.

*Strong Indistinguishability is proposed by (Derler et al., 2019), which defines higher security of chameleon hash.

*w-KE Security Model is introduced by (Huang et al., 2021), comparing with EUF-CMA security model, this model allows adversary corrupt all users and modifiers.

*Denote SIS as small integer solution problem and ISIS as inhomogeneous small integer solution problem.

However, there are some common disadvantages, including trapdoor key management and coarse-grained redaction, and other limitations. First, this method can not solve the problems caused by immutability in the public blockchain. In the public blockchain, the environment is entirely decentralized. Certain measures should be taken to ensure the security of the key sharing and the efficiency of the MPC protocol. Then, redaction only requires a consensus among trapdoor shareowners, which may make the blockchain suffer from a collusion attack. Furthermore, only the latest state of data is stored on the blockchain and the redacting process is transparent to the user. It makes the malicious redaction without any responsibility and data owners cannot recover the maliciously modified data.

To address the above problems, several researches [31], [32], [33] addressed the coarse-grained redaction and accountability separately by employing some cryptographic primitives. However, trapdoor key management is still a direction to be studied. One possible method is to adopt the same method of the committee in [52]. The trapdoor key is shared among some randomly selected committee members based on their stake or computing power.

Polynomial-based blockchain and RSA-based blockchain are based on other puzzles. They adopt a new way to link the block and ensure the stability of the blockchain. However, these methods face some new problems. First, creating a new block does not need too much computing power and the difficulty of block creation cannot be adjusted. Therefore, the PoW consensus algorithm is inappropriate in these structures. Second, these structures remedy defects of hash-based blockchain, but they suffer from some other attacks as mentioned in [39]. Finally, it can not provide public accountability and verifiability because the most recent one covers all the historical data.

Table V summarizes those schemes in Section IV in terms of their main idea and disadvantages. Voting-based redaction and local redaction are easy to integrate with existing blockchains and achieve accountability. Moreover, they do not involve complex cryptographic primitives.

However, non-cryptography-based methods face some new security issues. For data-appending-based redaction, the malicious user can specify the redaction policy so that only himself

can mute the active transaction. It also suffers from scalability issues when being performed in the public blockchain. For voting-based redaction, Dousti et al. [39] proposed the vote erasure attack, which makes the ledger inconsistent by erasing the vote on the blockchain. The redaction policy is essential to the security of the blockchain. Through the policy, some malicious behaviors can be prohibited. For example, it is easy to prevent the vote erasure attack by setting the condition that the proposed redaction does not redact votes for other redactions in the blockchain. Therefore, to protect the state of the blockchain, redaction should only be performed under the policy. The hard fork is not suitable for redaction in the blockchain. On the one hand, it is difficult for users to reach a consensus in the system. On the other hand, a hard fork requires computing a new blockchain, and all data still exists in the old blockchain, which would waste a lot of storage and computing resources.

In some schemes [50], [51], [52], voting is the key way to get all members to reach a consensus and ensure fairness of redaction. In Deuber et al.'s scheme [50], the voting protocol is based on the PoW consensus. Every miner has one vote in the block they mine. Marsalek and Zefferer's scheme [51] uses a similar approach to voting. However, the voting period is too long, which makes the schemes inefficient. Therefore, new voting protocols are necessary to achieve the scheme's efficiency.

As summarized above, we can obtain that these efforts are focused on the following aspects:

- Replacing the hash function with the chameleon hash function to find the collision of the block and transaction.
- Modifying the protocol so the block can be verified after redaction.
- Designing novel blockchain structures that support redaction easily.

VI. FUTURE RESEARCH DIRECTIONS

We posited that redactable blockchain-based applications or solutions would be increasingly mainstream in the foreseeable future, particularly in privacy-conscious jurisdictions. Our

TABLE V
SUMMARY OF REDACTABLE METHODS IN SECTION IV

Works	Main Idea	Main Disadvantages	Consensus Approach
Puddu et al. [49]	Transaction is represented by a transaction set with only one active transaction in the set. Redacting is achieved by modifying the active transaction.	1. Heavy storage overhead 2. No support for deleting 3. Scalability issues	Mutability policy
Deuber et al. [50]	Maintain two hash links simultaneously, one for verification and another for redaction.	1. Voting period is too long 2. No support for redacting stateful contents	Vote
Marsalek and Zefferer [51]	Maintain two chains simultaneously, one for verification and another for redaction.	1. Heavy storage overhead 2. Voting period is too long 3. Not backward compatible with existing blockchain	Vote
Florian et al. [56]	Users operate on their local ledger.	1. It is difficult to reach a global consensus 2. Many miners keep an incomplete ledger	—
Hard fork [57]	Users update the blockchain protocol.	1. It is difficult to reach consensus 2. Heavy storage and computation overhead	—

*Consensus approach refers to the consensus approach on redacting decisions.

survey of existing redaction mechanisms identified a number of limitations and challenges. It can also be observed from our survey that blockchain redaction technologies are generally not quite mature, partly evidenced by the lack of real-world implementations of proposed approaches. Here, we will briefly outline several potential research directions.

A. Security Issues

Some current schemes modify the blockchain's structure and protocol, bringing some potential security problems. As mentioned in the previous sections, RSA-based redaction may suffer from the moderator circumvention attack and the reversion attack. Chameleon-hash-based schemes are subject to collusion attacks and a single point of failure. It may be a future research direction by combining with other cryptographic primitives. For example, Xu et al. [33] mitigates malicious behavior through digital signatures. Moreover, some schemes use ABE for key sharing. In these schemes, the authorized user can make multiple edits to the blockchain and may even share the key with users who are not permitted. These problems can be solved by key revocation [63], [64], [65]. Key revocation reduces the loss of key leaks and updates user's access rights. Therefore, it could be a future research direction.

B. Security vs Efficiency

In chameleon-hash-based schemes, trapdoor key sharing impacts efficiency and security. If too many parties are involved, the redaction will not be efficient. If there are too few parties, the system relies on an honest majority to avoid collusion attacks (i.e., malicious users with redacting rights

will arbitrarily tamper with the data on the blockchain). For voting-based schemes such as those presented in [50], [51], and [52], if voting is performed on the blockchain then the voting period will be too long. In Bitcoin, for example, it takes about seven days to complete a vote. If community members vote off-chain, the process is much more efficient. However, such an approach relies on the assumption that most members are honest. Therefore, balancing security and efficiency in redacting is another potential research direction.

C. Stateful Redaction

In the Bitcoin system, data is usually inserted into the following three fields: Coinbase, OP_RETURN and Locking-Script. Among them, Coinbase and OP_RETURN are context-independent. Another challenge is that most existing schemes only support the redaction of context-independent content. However, redaction may cause inconsistencies in these schemes. If an erroneous transaction is redacted, subsequent transactions using it as the input will also be affected. For Ethereum, these schemes cannot be used to repair contracts that have a cascading effect. One solution is to refactor all subsequent transactions, and ensure that the number of digital currencies remains unchanged during the reconstruction process. Therefore, to ensure the stability of the blockchain state, supporting the redacting of state data may be the critical technology for blockchain redaction in future research.

D. Compatibility

Several existing schemes have modified block structures. Consequently such methods are incompatible with mainstream blockchain-based applications (e.g., Bitcoin and Ethereum).

TABLE VI
SUMMARY OF REDACTING METHODOLOGIES IN SECTIONS III AND IV

	[27]	[32]	[31]	[35]	[36]	[49]	[50]	[51]	[52]	[55]	[56]	[37]
Accountability	×	×	×	✓	✓	✓	✓	✓	✓	✓	—	×
MPC	✓	✓	✓	✓	×	×	×	×	×	×	×	×
Redaction Type	Global	Global	Global	Global	Global	Global	Global	Global	Global	Global	Local	Global
UTXO Set	×	×	×	×	×	×	×	✓	×	✓	✓	×
Penalty	×	×	×	×	✓	×	×	×	×	×	×	×
Existing Data	×	×	×	×	×	×	×	×	×	✓	✓	✓
Granularity	block	block	tx	block	block	tx	block	block	block	block	tx	block
Efficiency	high	high	high	middle	high	middle	low	low	high	low	middle	middle
Scalability	×	×	×	×	×	×	✓	✓	✓	✓	×	✓
Consensus	PoW	PoW	PoW	PoW	PoW	PoW	PoW	PoW	PoW	PoW, PoS	All Consensus	PBFT

*Accountability means the person who redacts the blockchain can be found.

*MPC means whether MPC protocol is required.

*UTXO Set means whether the UTXO Set can be redacted.

*Penalty means whether malicious modification will be punished.

*Granularity means redacting granularity of the method. Block means block-level, tx means transaction-level.

*Existing Data means whether the existing data can be redacted.

One solution is to refactor the blockchain with the new structure, which will incur significant computing resources, costs, and time. Solutions such as those of Thyagarajan [55] (proposed Reparo protocol performs as a publicly verifiable layer on the blockchain) may provide some insight into the future design of redaction schemes, for example, by modifying upper-level protocols instead of changing the blockchain's structure.

E. Universality

Most existing methods are designed for the Bitcoin system, since they modify the Bitcoin data structure and the underlying protocol. Consequently, these methods may not be applicable to other blockchain-based systems. In addition, some approaches are only designed for some specific consensus protocol (e.g., PoW). The polynomial based blockchain, for example, is only suitable for cooperative consensus algorithms. Therefore, there is a need to also focus on designing universal redacting techniques.

F. Applicability

Blockchain redaction technologies are not yet mature. Most of them are theoretical discussions. Most of schemes have no mature systems or applications to verify their effectiveness and security. Moreover, in order to comply with current laws and regulations, designing GDPR-compliant applications is an important current research direction. For example, Truong and Lee [66] discussed the recent research efforts of GDPR-compliant personal data management. Besides, redactable blockchain can address the security demands of many blockchain application scenarios, such as smart grids [67] and meta-verse [68]. Therefore, designing blockchain applications that support redacting is an important direction in the future.

G. Scalability

As shown in Table VI, secret sharing and MPC are used in some schemes. The complex cryptographic primitives will incur significant computation and communication overheads. For example, in [27], the trapdoor key will be shared among all parties. When redaction is performed, all parties need to cooperate to calculate the collision of the hash value. Computation time and communication consumption will increase as the number of participants increases. It can also impact the scalability of the schemes. Therefore, one possible direction is to improve these schemes by avoiding (or replacing) the use of large-scale MPC and key management.

VII. CONCLUSION

Recent years have witnessed the growing popularity of blockchain due to its decentralization and immutability. It provides a new way to store and share data in an untrusted environment. However, immutability has caused problems in some cases, which have hindered the application of blockchain. Therefore, to solve these problems, redactable blockchain has been an emerging topic in recent years.

In this paper, we presented a comprehensive analysis of redaction mechanisms. Then, we summarised and compared these methods. Furthermore, we identified future research directions for blockchain redaction.

REFERENCES

- [1] A. Group, "Antchain baas platform," 2016, Accessed: Apr. 18, 2022. [Online]. Available: <https://antchain.antgroup.com/>
- [2] Citibank, "blockchain-based open financing platform," 1998, Accessed: Apr. 18, 2022. [Online]. Available: <https://www.citibank.com/tts/insights/articles/article32.html>
- [3] J. Abou Jaoude and R. George Saade, "Blockchain applications—usage in different domains," *IEEE Access*, vol. 7, pp. 45360–45381, 2019.

- [4] L. Carter and J. Ubacht, "Blockchain applications in government," in *Proc. 19th Annu. Int. Conf. Digit. Government Res.: Governance Data Age*, Delft, The Netherlands, 2018, pp. 1–2.
- [5] D. Dujak and D. Sajter, "Blockchain applications in supply chain," in *SMART Supply Network*, Berlin, Germany: Springer, 2019, pp. 21–46.
- [6] M. Schellekens, "Does regulation of illegal content need reconsideration in light of blockchains?," *Int. J. Law Inf. Technol.*, vol. 27, no. 3, pp. 292–305, 2019.
- [7] I. of Business, "Bitcoin blockchain contains porn, say researchers. not news, say coders," 2019, Accessed: Apr. 18, 2022. [Online]. Available: <https://internetofbusiness.com/bitcoin-blockchain-contains-illegal-porn-say-researchers/>
- [8] E. U. Regulation, "General data protection regulation," 2016, Accessed: Apr. 18, 2022. [Online]. Available: <https://gdpr-info.eu/>
- [9] S. C. of the National People's Congress, "Personal information protection law," 2021, Accessed: Apr. 18, 2022. [Online]. Available: <http://www.npc.gov.cn/npc/c30834/202108/a8c4e3672c74491a80b53a172bb753fe.shtml>
- [10] A. Carvalho, J. W. Merhout, Y. Kadiyala, and J. Bentley II, "When good blocks go bad: Managing unwanted blockchain data," *Int. J. Inf. Manage.*, vol. 57, 2021, Art. no. 102263.
- [11] S. Sayeed, H. Marco-Gisbert, and T. Caira, "Smart contract: Attacks and protections," *IEEE Access*, vol. 8, pp. 24416–24427, 2020.
- [12] M. I. Mehar et al., "Understanding a revolutionary and flawed grand experiment in blockchain: The dao attack," *J. Cases Inf. Technol.*, vol. 21, no. 1, pp. 19–32, 2019.
- [13] N. Atzei, M. Bartoletti, and T. Cimoli, "A survey of attacks on ethereum smart contracts (SoK)," in *Proc. Int. Conf. Princ. Secur. Trust*, 2017, pp. 164–186.
- [14] Statista, "Size of the bitcoin blockchain from january 2009 to april 4, 2022," 2022. Accessed: Apr. 18, 2022. [Online]. Available: <http://www.npc.gov.cn/npc/c30834/202108/a8c4e3672c74491a80b53a172bb753fe.shtml>
- [15] E. Politou, F. Casino, E. Alepis, and C. Patsakis, "Blockchain mutability: Challenges and proposed solutions," *IEEE Trans. Emerg. Topics Comput.*, vol. 9, no. 4, pp. 1972–1986, Oct.-Dec. 2021.
- [16] Y. Yong and W. Feiyue, "Editable blockchain: Models, techniques and methods," *ACTA Automatica Sinica*, vol. 46, no. 5, pp. 831–846, 2020.
- [17] A. B. Haque, A. K. M. N. Islam, S. Hyrnsalmi, B. Naqvi, and K. Smolander, "GDPR compliant blockchains—a systematic literature review," *IEEE Access*, vol. 9, pp. 50593–50606, 2021.
- [18] C. Decker and R. Wattenhofer, "Bitcoin transaction malleability and mtgox," in *Proc. Eur. Symp. Res. Comput. Secur.*, Wroclaw, Poland, 2014, pp. 313–326.
- [19] R. C. Merkle, "A digital signature based on a conventional encryption function," in *Conf. Theory Application Cryptographic Techn.* pp. 369–378, 1987.
- [20] J. Garay, A. Kiayias, and N. Leonardos, "The bitcoin backbone protocol: Analysis and applications," in *Proc. Annu. Int. Conf. Theory Appl. Cryptographic Techn.*, 2015, pp. 281–310.
- [21] S. Nakamoto, "Bitcoin: A peer-to-peer electronic cash system," 2008. Accessed: Apr. 18, 2022. [Online]. Available: <https://bitcoin.org/bitcoin.pdf>
- [22] S. King and S. Nadal, "PPCoin: Peer-to-peer crypto-currency with proof-of-stake," *Self-Published Paper*, vol. 19, no. 1, pp. 1–6, Aug. 2012.
- [23] M. Castro et al., "Practical byzantine fault tolerance," in *Proc. 3rd Symp. Operating Syst. Des. Implementation*, 1999, pp. 173–186.
- [24] Y. Wang, Z. Su, J. Ni, N. Zhang, and X. Shen, "Blockchain-empowered space-air-ground integrated networks: Opportunities, challenges, and solutions," *IEEE Commun. Surveys Tuts.*, vol. 24, no. 1, pp. 160–209, Jan.-Mar. 2021.
- [25] H. Krawczyk and T. Rabin, "Chameleon hashing and signatures," 1997. [Online]. Available: <http://wwwresearch.ibm.com/security/projects.html>
- [26] G. Ateniese and B. de Medeiros, "On the key exposure problem in chameleon hashes," in *Proc. Int. Conf. Secur. Commun. Netw.*, 2004, pp. 165–179.
- [27] G. Ateniese, B. Magri, D. Venturi, and E. Andrade, "Redactable blockchain— or— rewriting history in bitcoin and friends," in *Proc. IEEE Eur. Symp. Secur. Privacy*, 2017, pp. 111–126.
- [28] D. Naccache, D. Pointcheval, and J. Stern, "Twin signatures: An alternative to the hash-and-sign paradigm," in *Proc. 8th ACM Conf. Comput. Commun. Secur.*, 2001, pp. 20–27.
- [29] Accenture and D. G. Ateniese, "Accenture debuts prototype of 'editable' blockchain for enterprise and permissioned systems," 2016, Accessed: Apr. 18, 2022. [Online]. Available: <https://www.ccn.com/accenture-unveil-editable-blockchain-prototype/>
- [30] E. Androulaki et al., "Hyperledger fabric: A distributed operating system for permissioned blockchains," in *Proc. 13th EuroSys Conf.*, 2018, pp. 1–15.
- [31] D. Derler, K. Samelin, D. Slamanig, and C. Striecks, "Fine-grained and controlled rewriting in blockchains: Chameleon-hashing gone attribute-based," *IACR Cryptol. ePrint Arch.*, vol. 2019, 2019, Art. no. 406.
- [32] K. Huang et al., "Building redactable consortium blockchain for industrial Internet-of-Things," *IEEE Trans. Ind. Inform.*, vol. 15, no. 6, pp. 3670–3679, Jun. 2019.
- [33] S. Xu, J. Ning, J. Ma, X. Huang, and R. H. Deng, "K-time modifiable and epoch-based redactable blockchain," *IEEE Trans. Inf. Forensics Secur.*, vol. 16, pp. 4507–4520, 2021.
- [34] C. Wu, L. Ke, and Y. Du, "Quantum resistant key-exposure free chameleon hash and applications in redactable blockchain," *Inf. Sci.*, vol. 548, pp. 438–449, 2021.
- [35] J. Camenisch, D. Derler, S. Krenn, H. C. Pöhls, K. Samelin, and D. Slamanig, "Chameleon-hashes with ephemeral trapdoors," in *Proc. IACR Int. Workshop Public Key Cryptogr.*, 2017, pp. 152–182.
- [36] J. Ma, S. Xu, J. Ning, X. Huang, and R. H. Deng, "Redactable blockchain in decentralized setting," *IEEE Trans. Inf. Forensics Secur.*, vol. 17, pp. 1227–1242, 2022.
- [37] L. Cheng, J. Liu, C. Su, K. Liang, G. Xu, and W. Wang, "Polynomial-based modifiable blockchain structure for removing fraud transactions," *Future Gener. Comput. Syst.*, vol. 99, pp. 154–163, 2019.
- [38] D. Grigoriev and V. Shpilrain, "RSA and redactable blockchains," *Int. J. Comput. Math.: Comput. Syst. Theory*, vol. 6, no. 1, pp. 1–6, 2021.
- [39] M. S. Dousti and A. Küpcü, "Moderated redactable blockchains: A definitional framework with an efficient construct," in *Proc. Data Privacy Manage., Cryptocurrencies Blockchain Technol.*, pp. 355–373.
- [40] A. K. Kasgar, J. Agrawal, and S. Shahu, "New modified 256-bit MD 5 algorithm with SHA compression function," *Int. J. Comput. Appl.*, vol. 42, no. 12, pp. 47–51, 2012.
- [41] O. Goldreich, "Secure multi-party computation," *Manuscript Preliminary Version*, vol. 78, p. 110, 1998.
- [42] A. Shamir, "How to share a secret," *Commun. ACM*, vol. 22, no. 11, pp. 612–613, 1979.
- [43] C. Zhang, Z. Ni, Y. Xu, E. Luo, L. Chen, and Y. Zhang, "A trustworthy industrial data management scheme based on redactable blockchain," *J. Parallel Distrib. Comput.*, vol. 152, pp. 167–176, 2021.
- [44] J. Xu, K. Xue, H. Tian, J. Hong, D. S. L. Wei, and P. Hong, "An identity management and authentication scheme based on redactable blockchain for mobile networks," *IEEE Trans. Veh. Technol.*, vol. 69, no. 6, pp. 6688–6698, Jun. 2020.
- [45] R. Mishra, D. Ramesh, and N. Mohammad, "RBDA: Redactable-blockchain based secure data aggregation scheme for IoT enabled cloud paradigm," in *Proc. IEEE Int. Conf. Pervasive Comput. Commun. Workshops Other Affiliated Events*, 2022, pp. 409–414.
- [46] J. Wei et al., "A redactable blockchain framework for secure federated learning in industrial Internet-of-Things," *IEEE Internet Things J.*, vol. 9, no. 18, pp. 17901–17911, Sep. 2022.
- [47] E. Meijering, "A chronology of interpolation: From ancient astronomy to modern signal and image processing," *Proc. IEEE*, vol. 90, no. 3, pp. 319–342, Mar. 2002.
- [48] R. L. Rivest, A. Shamir, and L. Adleman, "A method for obtaining digital signatures and public-key cryptosystems," *Commun. ACM*, vol. 21, no. 2, pp. 120–126, 1978.
- [49] I. Puddu, A. Dmitrienko, and S. Capkun, "μchain: How to forget without hard forks," *IACR Cryptol. ePrint Arch., Paper 2017/106*, 2017. [Online]. Available: <https://eprint.iacr.org/2017/106>
- [50] D. Deuber, B. Magri, and S. A. K. Thyagarajan, "Redactable blockchain in the permissionless setting," in *Proc. IEEE Symp. Secur. Privacy*, 2019, pp. 124–138.
- [51] A. Marsalek and T. Zeffere, "A correctable public blockchain," in *Proc. 18th IEEE Int. Conf. Trust. Secur. Privacy Comput. Commun./13th Int. Conf. Big Data Sci. Eng.*, 2019, pp. 554–561.
- [52] X.-Y. Li, J. Xu, L.-Y. Yin, Y. Lu, Q. Tang, and Z.-F. Zhang, "Escaping from consensus: Instantly redactable blockchain protocols in permissionless setting," *IEEE Trans. Dependable Secure Comput.*, pp. 1–20, 2022.
- [53] E. Daniel and F. Tschorsch, "Towards verifiable mutability for blockchains," 2021, *arXiv:2106.15935*.
- [54] J. Xu, X. Li, L. Yin, B. Guo, H. Feng, and Z. Zhang, "Redactable proof-of-stake blockchain with fast confirmation," *IACR Cryptol.*, vol. 2019, p. 1110, 2019.
- [55] S. A. K. Thyagarajan, A. Bhat, B. Magri, D. Tschudi, and A. Kate, "Reparo: Publicly verifiable layer to repair blockchains," in *Proc. Int. Conf. Financial Cryptography Data Secur.*, 2021, pp. 37–56.
- [56] M. Florian, S. Henningsen, S. Beaucamp, and B. Scheuermann, "Erasing data from blockchain nodes," in *Proc. IEEE Eur. Symp. Secur. Privacy Workshops*, 2019, pp. 367–376.

- [57] I.-C. Lin and T.-C. Liao, "A survey of blockchain security issues and challenges," *Int. J. Netw. Secur.*, vol. 19, no. 5, pp. 653–659, 2017.
- [58] J. Bruce, "The mini-blockchain scheme," White paper, vol. 1, pp. 1–10, 2014.
- [59] R. Matzutt, B. Kalde, J. Pennekamp, A. Drichel, M. Henze, and K. Wehrle, "How to securely prune bitcoin's blockchain," in *Proc. IFIP Netw. Conf.*, 2020, pp. 298–306.
- [60] N.-Y. Lee, J. Yang, M. M. H. Onik, and C.-S. Kim, "Modifiable public blockchains using truncated hashing and sidechains," *IEEE Access*, vol. 7, pp. 173571–173582, 2019.
- [61] X. Feng et al., "Pruneable sharding-based blockchain protocol," *Peer-to-Peer Netw. Appl.*, vol. 12, no. 4, pp. 934–950, 2019.
- [62] A. Chepur, M. Larangeira, and A. Ojiganov, "Rollerchain, a blockchain with safely pruneable full blocks," 2016, *arXiv:1603.07926*.
- [63] Y. Jia, S.-F. Sun, Y. Zhang, Z. Liu, and D. Gu, "Redactable blockchain supporting supervision and self-management," in *Proc. ACM Asia Conf. Comput. Commun. Secur.*, 2021, pp. 844–858.
- [64] A. Sahai, H. Seyalioglu, and B. Waters, "Dynamic credentials and ciphertext delegation for attribute-based encryption," in *Proc. Annu. Cryptology Conf.*, 2012, pp. 199–217.
- [65] S. Xu, G. Yang, Y. Mu, and R. H. Deng, "Secure fine-grained access control and data sharing for dynamic groups in the cloud," *IEEE Trans. Inf. Forensics Secur.*, vol. 13, no. 8, pp. 2101–2113, Aug. 2018.
- [66] N. B. Truong, K. Sun, G. M. Lee, and Y. Guo, "GDPR-compliant personal data management: A blockchain-based solution," *IEEE Trans. Inf. Forensics Secur.*, vol. 15, pp. 1746–1761, 2020.
- [67] Y. Wang et al., "SPDS: A secure and auditable private data sharing scheme for smart grid based on blockchain," *IEEE Trans. Ind. Inform.*, vol. 17, no. 11, pp. 7688–7699, Nov. 2021.
- [68] Y. Wang et al., "A survey on metaverse: Fundamentals, security, and privacy," *IEEE Commun. Surveys Tuts.*, early access, Sep. 07, 2022, doi: [10.1109/COMST.2022.3202047](https://doi.org/10.1109/COMST.2022.3202047).



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